

CSE 340 Quiz-3 – Section 26

Full Marks: 20

Date: Dec 22, 2025

Time: 30 minutes

Name: _____

Roll: _____

1. Complete the following table with control signals for each given instruction. Put “x” for any unused control signal. [5 marks]

Instruction	RegWrite	MemRead	MemWrite	Branch	MemtoReg
Bne X3,X4, Else	0	0	0	1	x
Ld X3, 32(X4)	1	1	0	0	1
Sd X3, 32(X4)	0	0	1	0	0
Addi x5, x6, 9	1	0	0	0	0

2. Based on the value of (your ID%3), draw the datapath of the following instruction. [12 marks]

0 → Store, 1 → Load, 2 → Beq

You have this in the slides.

3. Why do we need a unit called “ALU control”? Why can’t we use the “Control” unit for this purpose? [3 marks]

Because multiple instructions often share the same ALU operation (such as load, store, and addi). Therefore, we cannot determine the exact operation type solely from the opcode. That's why we need the ALU control that gets 2 bits from the original control unit and also uses func3 and some other bits to determine the exact ALU operation that needs to be done.